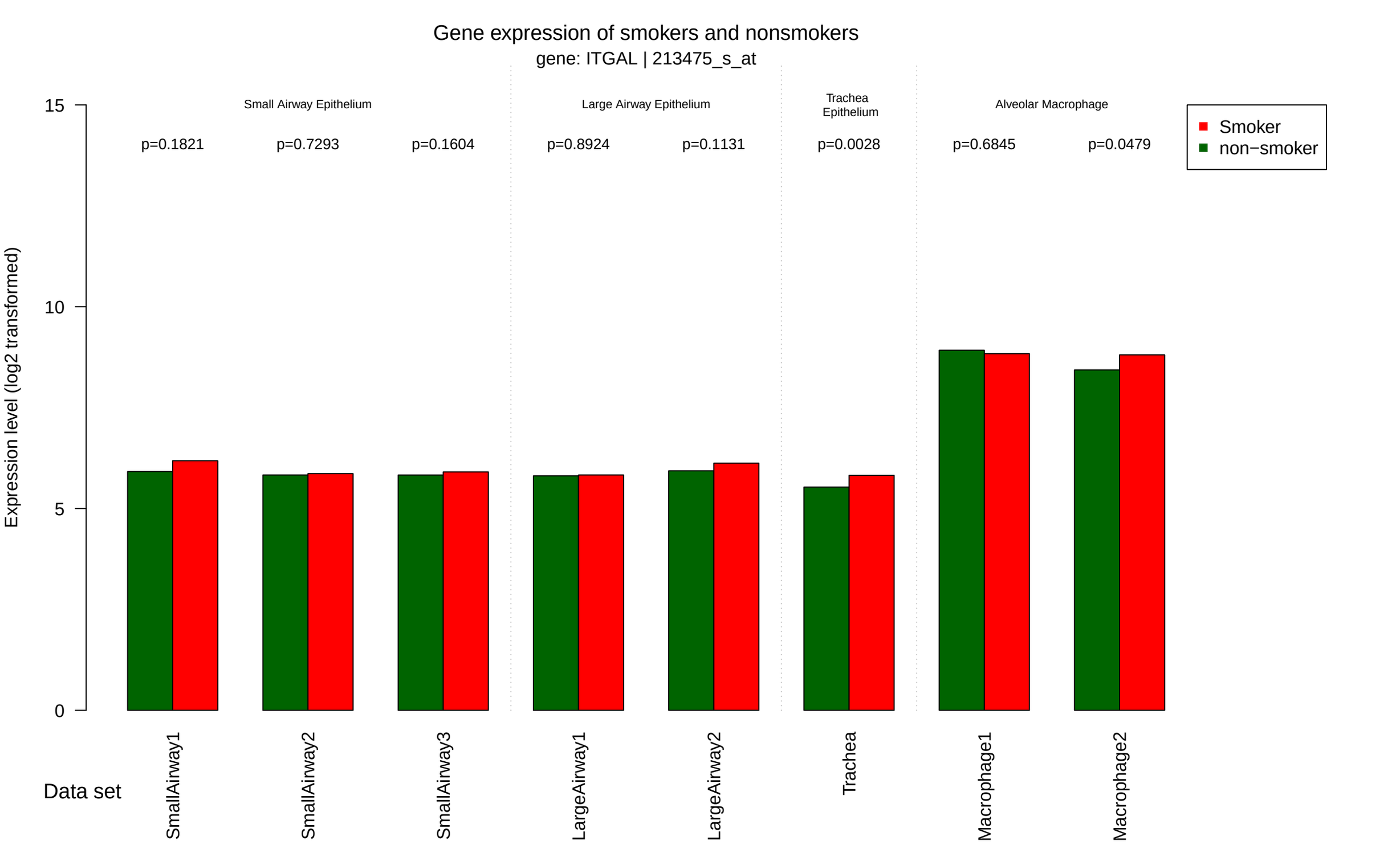
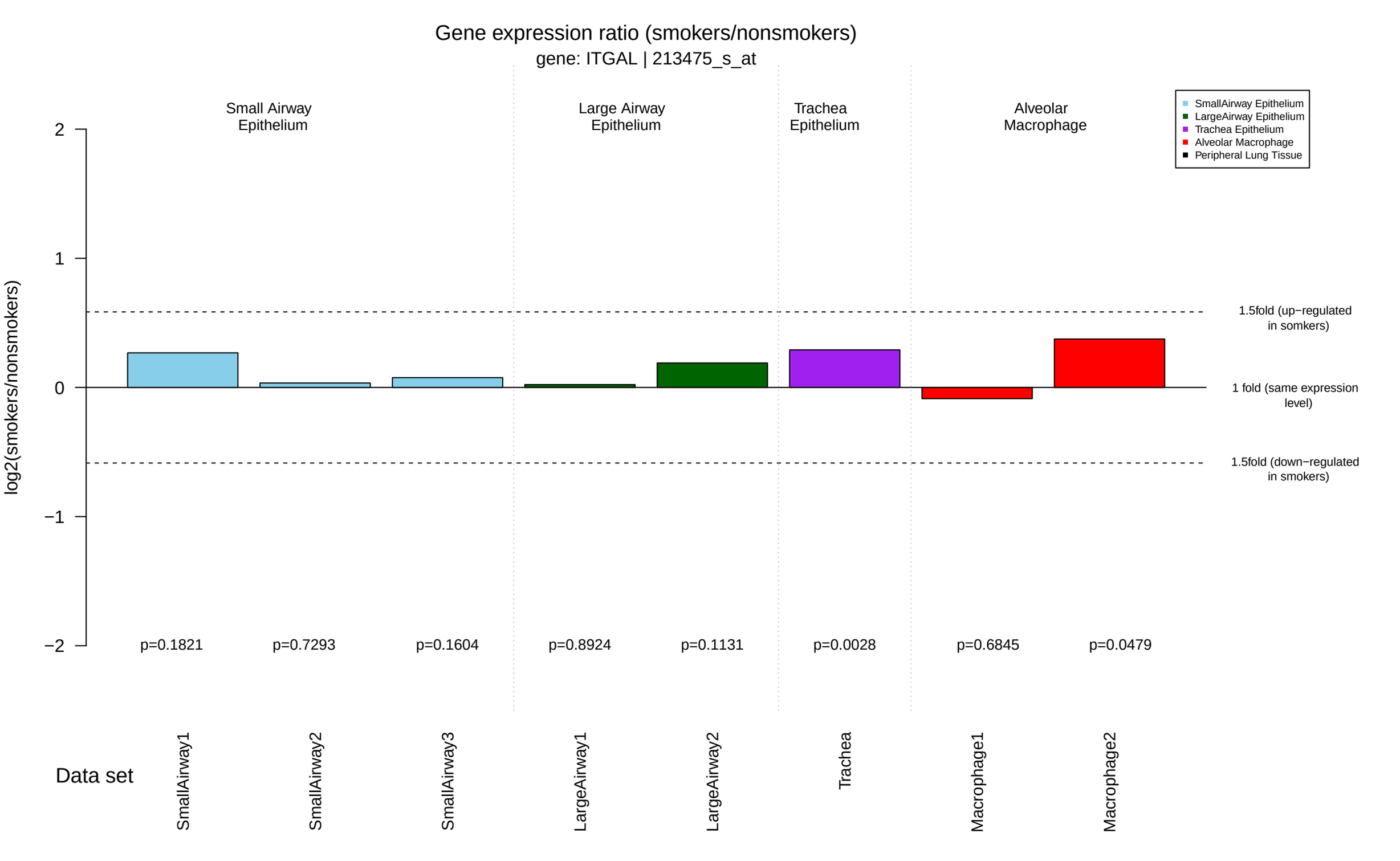
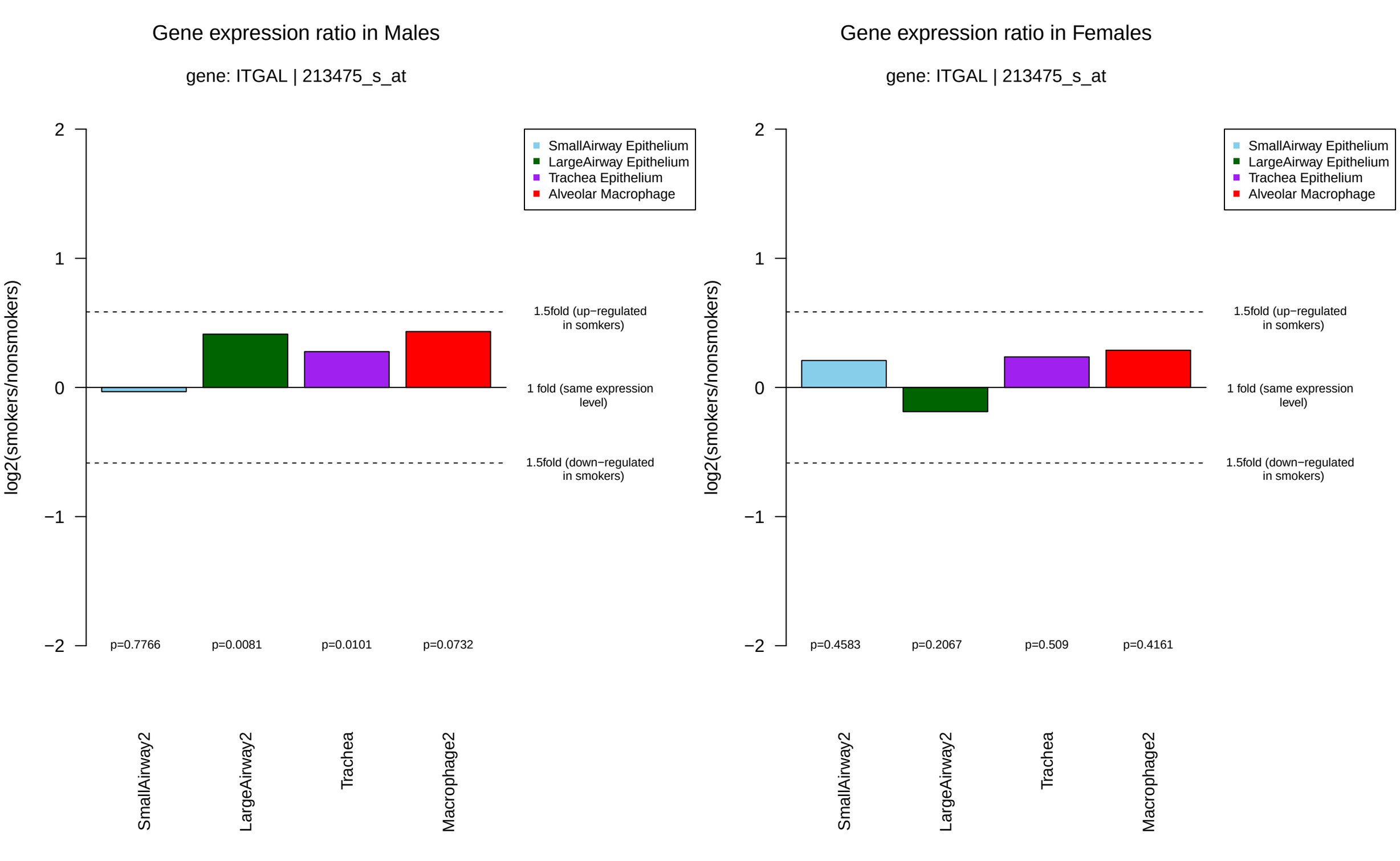
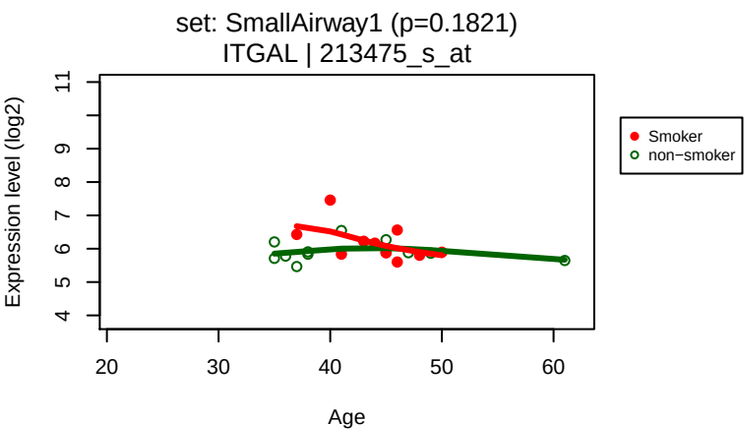
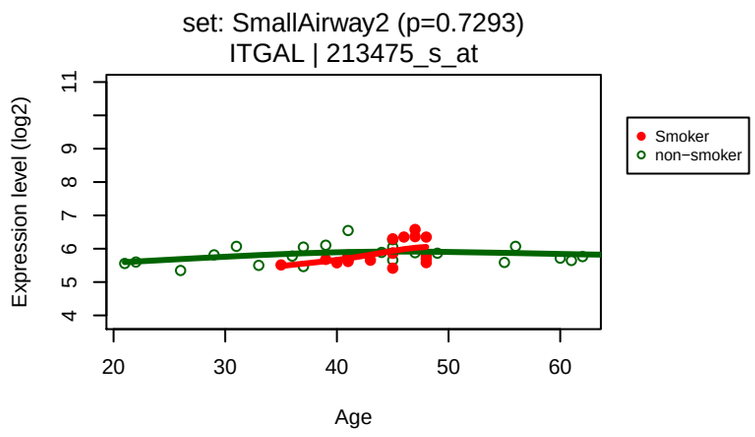
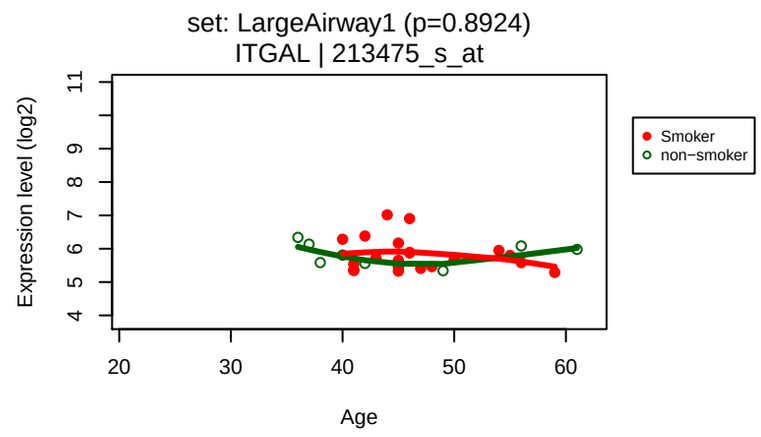

Gene expression of smokers and nonsmokers
gene: ITGAL | 213475_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1821
p=0.7293
p=0.1604
p=0.8924
p=0.1131
p=0.0028
p=0.6845
p=0.0479
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

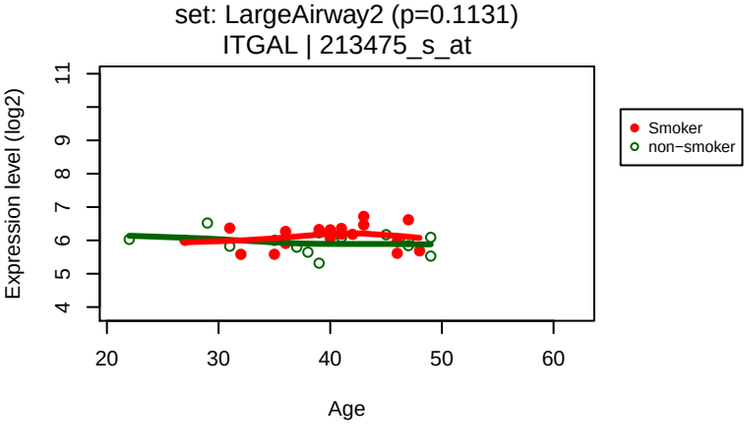

Gene expression ratio (smokers/nonsmokers)
gene: ITGAL | 213475_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1821
p=0.7293
p=0.1604
p=0.8924
p=0.1131
p=0.0028
p=0.6845
p=0.0479
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

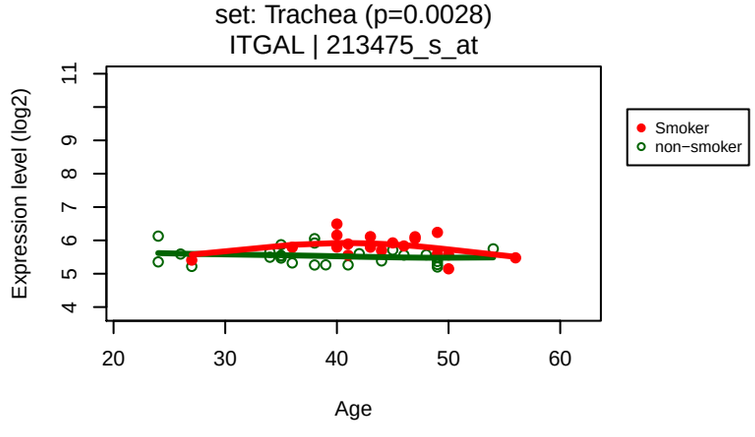

Gene expression ratio in Males
gene: ITGAL | 213475_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7766
p=0.0081
p=0.0101
p=0.0732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITGAL | 213475_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4583
p=0.2067
p=0.509
p=0.4161
SmallAirway2
LargeAirway2
Trachea
Macrophage2

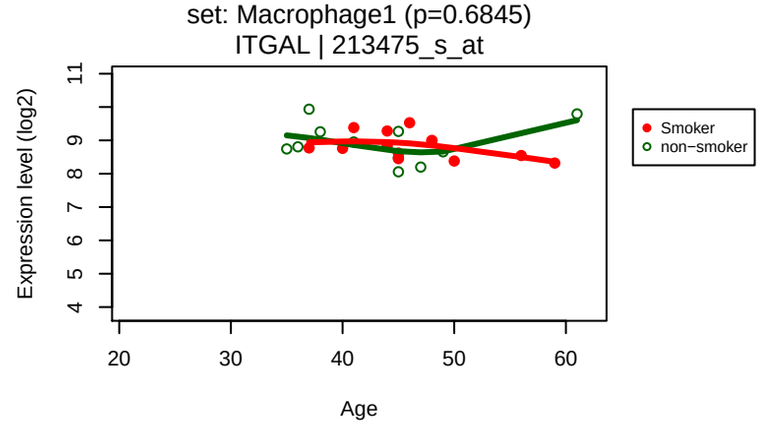

set: SmallAirway1 (p=0.1821)
ITGAL | 213475_s_at
Expression level (log2)
Age
Smoker
non−smoker

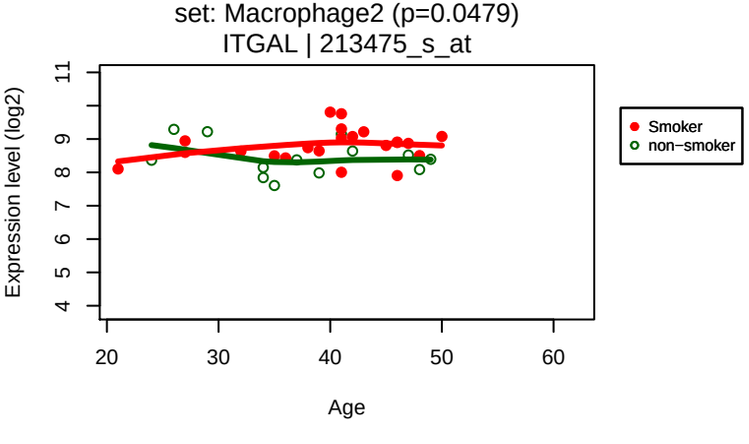

set: SmallAirway2 (p=0.7293)
ITGAL | 213475_s_at
Expression level (log2)
Age
Smoker
non−smoker

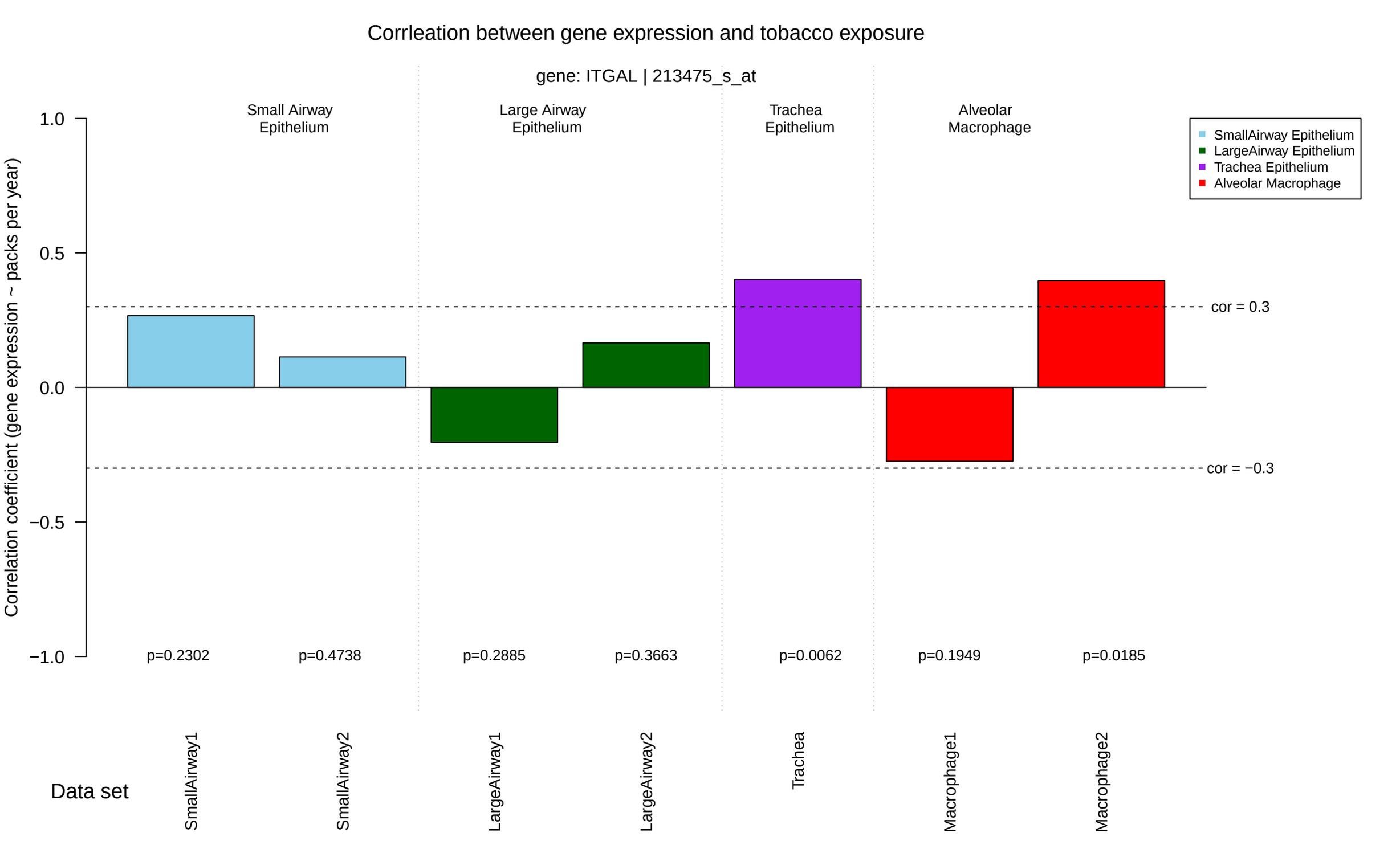

set: LargeAirway1 (p=0.8924)
ITGAL | 213475_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1131)
ITGAL | 213475_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0028)
ITGAL | 213475_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6845)
ITGAL | 213475_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0479)
ITGAL | 213475_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITGAL | 213475_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2302
p=0.4738
p=0.2885
p=0.3663
p=0.0062
p=0.1949
p=0.0185
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2